

Name: Peram Mahitha

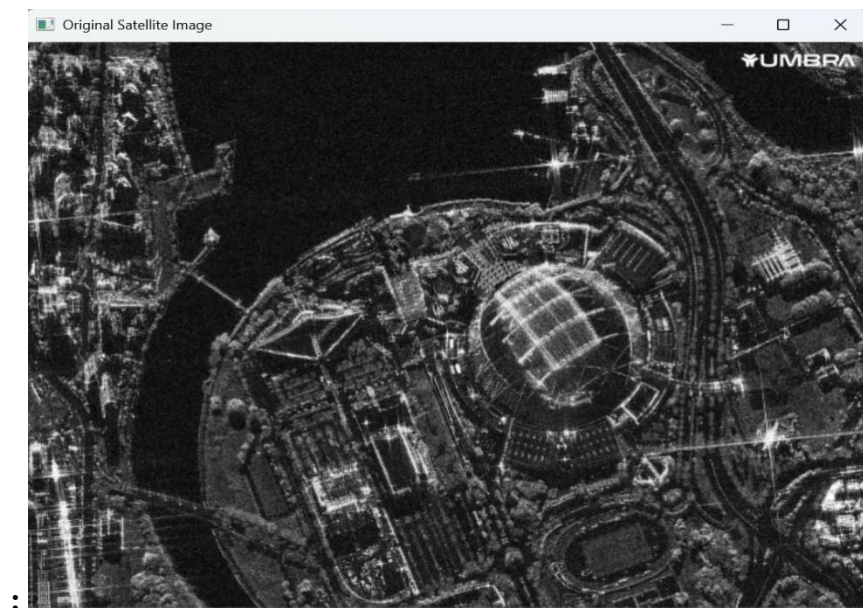
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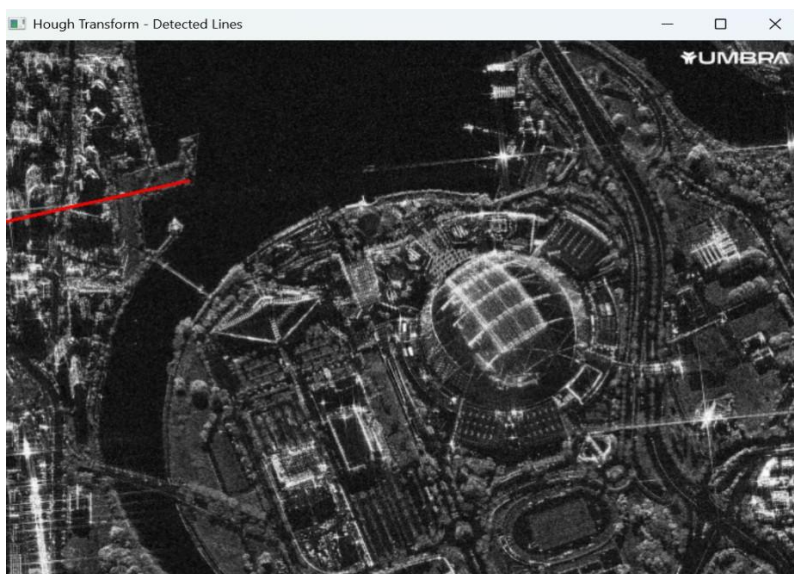
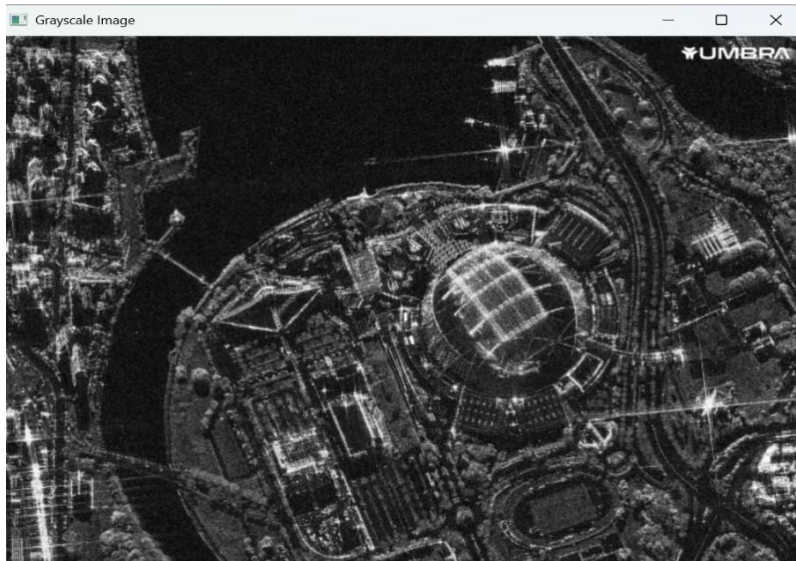
Question 1: Hough Transform & Shape Detection

Algorithm:

1. Opens a file-selection window.
2. You select a PNG/JPG/TIFF satellite image.
3. Converts it to grayscale.
4. Reduces noise using Gaussian Blur.
5. Detects edges using Canny Edge Detection.
6. Applies Hough Line Transform.
7. Draws detected roads/rivers/linear features on the original image.
8. Displays the results.

Output:



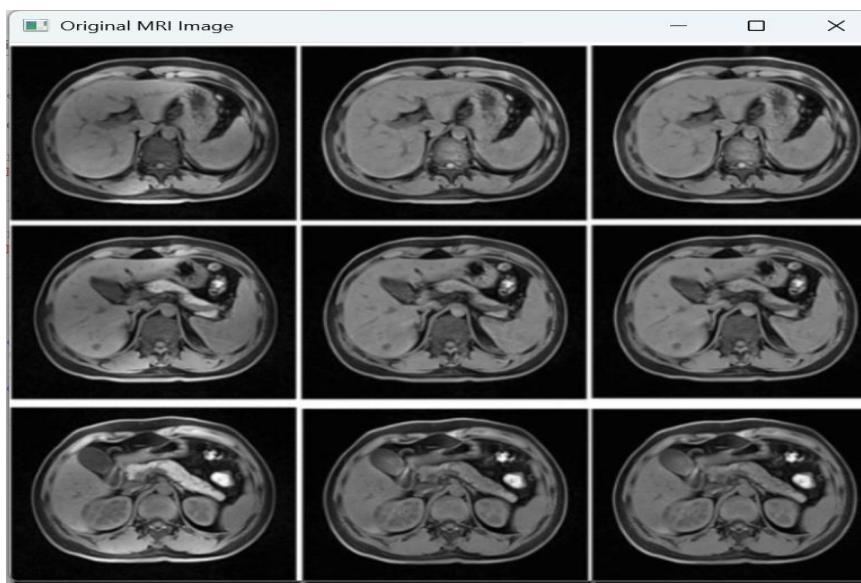


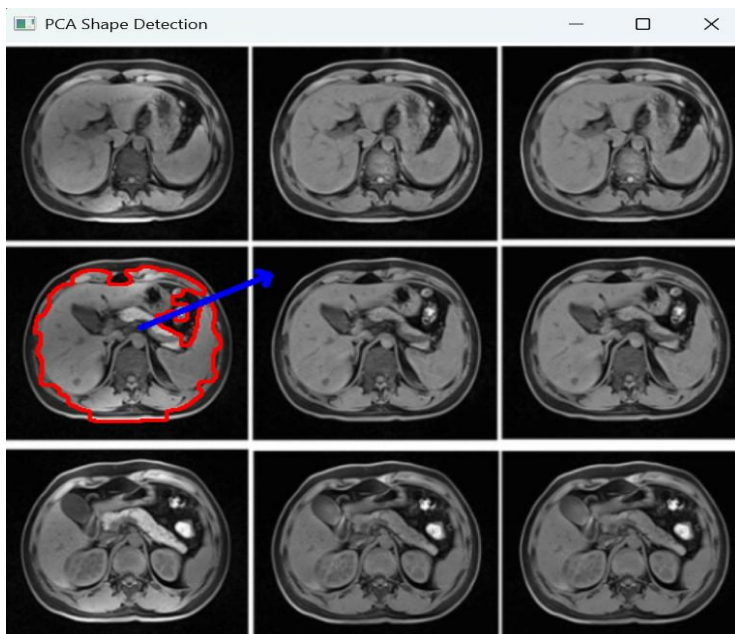
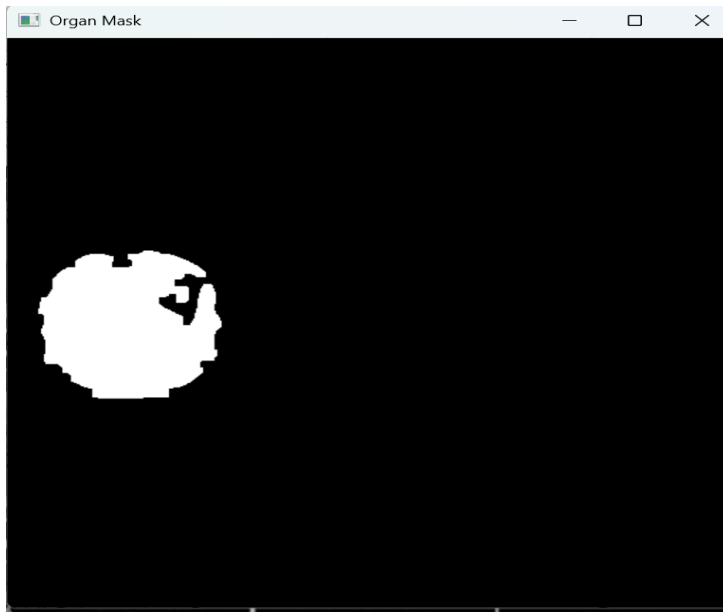
Question 2: PCA & Shape Priors for Object Recognition

Algorithm:

1. Start
2. Select one MRI image from the computer.
3. Read the MRI image in grayscale.
4. Resize the image to 512×512 pixels.
5. Apply Gaussian Blur to reduce noise.
6. Apply Otsu Thresholding to separate the organ from the background.
7. Apply morphological operations to remove small noise and fill gaps.
8. Detect the contours in the binary image.
9. Select the largest contour as the possible organ.
10. Extract the contour points of the detected organ.
11. Calculate the mean shape position using the contour points.
12. Calculate the covariance matrix of the centered points.
13. Perform PCA to obtain eigenvalues and eigenvectors.
14. Select the principal eigenvector to determine the main shape direction.
15. Draw the detected organ boundary and PCA principal axis on the MRI image.
16. Display the original MRI, organ mask, and PCA result.
17. Save the segmented organ and PCA result.
18. Stop.

Output:





Question 3: Machine Learning Methods (SVM, HMM, GMM, EM)

Algorithm:

1. Select your video
2. Resize each frame to a small size
3. Process only every 5th frame
4. Show progress

5. Extract optical-flow features
6. Predict the activity using SVM

Output:

```

IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help
Python 3.14.6 (tags/v3.14.6:c63a6c6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/ADMIN/OneDrive/Attachments/CO4 AT1 exp-3.py =====
Selected video:
C:/Users/ADMIN/Downloads/8484856-uhd_2160_3840_24fps.mp4
Total frames: 306
Processing frame: 5 / 306
Processing frame: 10 / 306
Processing frame: 15 / 306
Processing frame: 20 / 306
Processing frame: 25 / 306
Processing frame: 30 / 306
Processing frame: 35 / 306
Processing frame: 40 / 306
Processing frame: 45 / 306
Processing frame: 50 / 306
Processing frame: 55 / 306
Processing frame: 60 / 306
Processing frame: 65 / 306
Processing frame: 70 / 306
Processing frame: 75 / 306
Processing frame: 80 / 306
Processing frame: 85 / 306
Processing frame: 90 / 306
Processing frame: 95 / 306
Processing frame: 100 / 306
Processing frame: 105 / 306
Processing frame: 110 / 306
Processing frame: 115 / 306
Processing frame: 120 / 306
Processing frame: 125 / 306
Processing frame: 130 / 306
Processing frame: 135 / 306
Processing frame: 140 / 306
Processing frame: 145 / 306
Processing frame: 150 / 306
Processing frame: 155 / 306
Processing frame: 160 / 306
Processing frame: 165 / 306
Processing frame: 170 / 306

```

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IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help
Processing frame: 175 / 306
Processing frame: 180 / 306
Processing frame: 185 / 306
Processing frame: 190 / 306
Processing frame: 195 / 306
Processing frame: 200 / 306
Processing frame: 205 / 306
Processing frame: 210 / 306
Processing frame: 215 / 306
Processing frame: 220 / 306
Processing frame: 225 / 306
Processing frame: 230 / 306
Processing frame: 235 / 306
Processing frame: 240 / 306
Processing frame: 245 / 306
Processing frame: 250 / 306
Processing frame: 255 / 306
Processing frame: 260 / 306
Processing frame: 265 / 306
Processing frame: 270 / 306
Processing frame: 275 / 306
Processing frame: 280 / 306
Processing frame: 285 / 306
Processing frame: 290 / 306
Processing frame: 295 / 306
Processing frame: 300 / 306
Processing frame: 305 / 306

Video processing completed!
Number of feature sets: 61

Extracted Features:
X Motion: 0.7820486
Y Motion: 0.29859692
Motion Magnitude: 6.142026

Training SVM...
SVM training completed!

=====
HUMAN ACTIVITY RECOGNITION
=====
Detected Activity: Running
=====

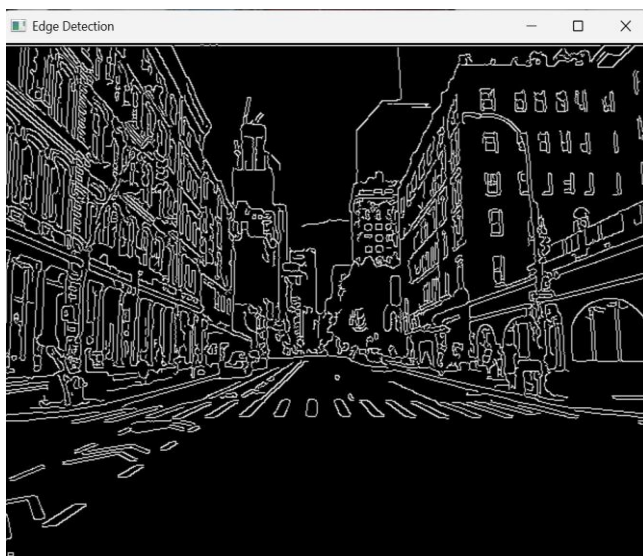
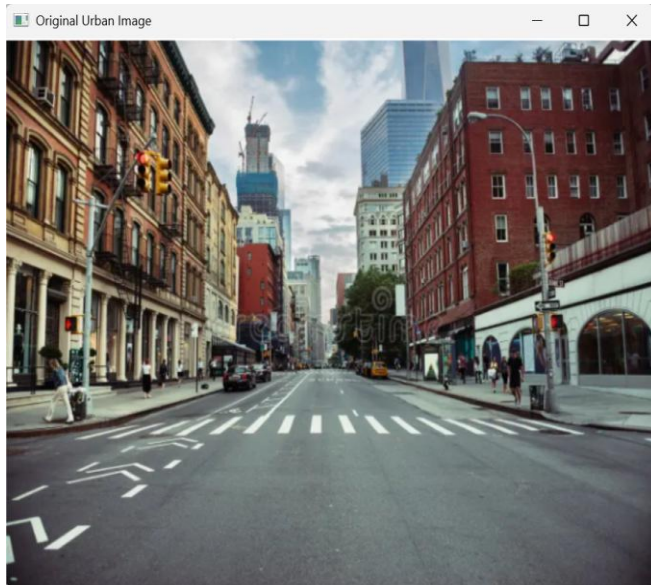
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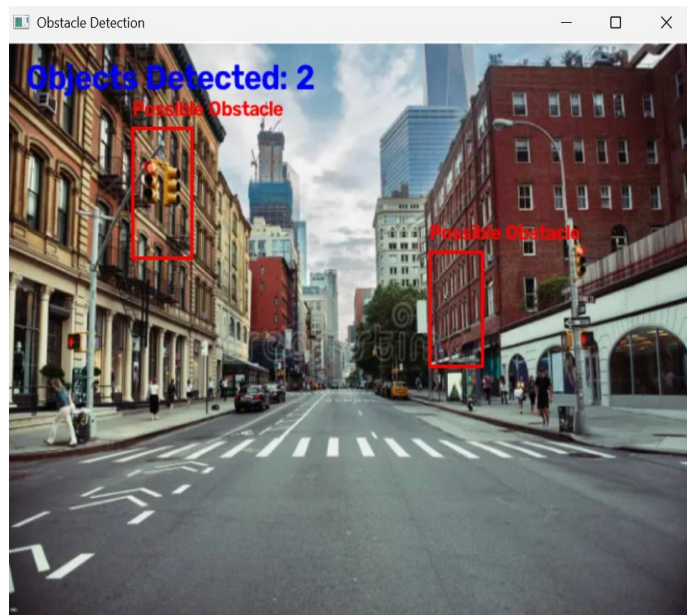
Question 4: Deep Learning (CNNs, Transformers, GANs)

Algorithm:

1. Let you select a JPG/PNG image.
2. Resizes it.
3. Builds a small CNN.
4. Trains on sample classes.
5. Predicts whether the image contains a Traffic Cone, Animal, or Other.

Output:





Question 5: Graph-Based Learning for Shape Correspondence

Algorithm:

1. let you select an urban road image from your files
2. detects objects/obstacles
3. draws bounding boxes
4. displays the result

Output:

